

A network interface card (NIC) is a hardware component, typically a circuit board or chip, installed on a computer so it can connect to a network. Modern NICs provide functionality to computers, such as support for input/output interrupt, direct-memory access interfaces, data transmission, network traffic engineering and partitioning.

A NIC provides a computer with a dedicated point of connectivity to a network. This network may be a LAN, wide area network ([WAN](#)) or even the public internet. NICs transmit and receive packets between the network and endpoint. They convert data from [analog to digital](#) (from the network to the device) and digital to analog (from the device to the network). They also manage data flows to control traffic and maintain optimal network performance.

To communicate on any type of network, a device -- also known as a *node* or *endpoint* -- must have a NIC installed. Most modern computers have built-in NICs, although NICs can also be inserted into devices' expansion slots to enable network connectivity.

NICs allow computers to [connect to a network](#) at much higher speeds than possible with legacy technologies such as dial-up modems. They also reduce network collisions by using algorithms and protocols. Some NICs also use error correction codes to detect and correct errors, thus enhancing data integrity.

How does a NIC work?

The Open Systems Interconnection ([OSI](#)) model helps describe how NICs work -- sending signals at the physical layer, transmitting data packets at the network layer and operating as an interface at the [TCP/IP](#) layer. In a computing device, a NIC contains the [physical layer](#) circuitry necessary for communicating with a data link layer standard, such as Ethernet or Wi-Fi. Each network interface card represents a device and can prepare, transmit and control the data flow on the network.

The OSI model

LAYER 7	APPLICATION Network process to application
LAYER 6	PRESENTATION Data representation and encryption
LAYER 5	SESSION Interhost communication
LAYER 4	TRANSPORT End-to-end connections and reliability
LAYER 3	NETWORK Path determination and IP
LAYER 2	DATA LINK MAC and LLC (Physical addressing)
LAYER 1	PHYSICAL Media, signal and binary transmission

The functions of a network interface card correlate to layers 1, 3 and 4 of the OSI model.

The NIC operates as a middleman between a computer and a data network. For example, when a user requests a webpage, the computer passes the request to the network card, which converts it into electrical impulses.

A web server on the internet receives these impulses and responds by sending the webpage back to the network card as electrical signals. The card receives these signals and translates them into the data that the computer displays. NICs use **unique MAC addresses** to identify devices on a network and ensure that data packets go to the correct device.

NICs were originally implemented as expansion cards that could plug into a computer port, router or USB device. However, more modern network cards are built directly into the computer's **motherboard** chipset. Users can purchase expansion card NICs online or in retail stores if additional independent network connections are needed. When

users choose a NIC, its specifications should correspond with the network standard.

Types of NICs

The standard NIC is a plastic circuit board that slides into a computer to connect with the motherboard. However, this connection can occur in multiple ways, which is why many types of NICs are now available.



can be installed on motherboards three different ways.

These include the following:

- **Wireless.** NICs that use an antenna provide wireless reception through radio frequency waves. The antenna communicates with a wireless access point via the RF waves. Wi-Fi connections use wireless NICs.
- **Wired.** These NICs have input jacks made for cables. Ethernet is the most popular wired LAN technology.
- **Fiber optics.** Some NICs use fiber optic cables to connect nodes to networks. These NICs may be used as a high-speed support system for network traffic handling on server computers. It's also possible to achieve this support by combining multiple NICs. Fiber optic NICs are typically more expensive and complex than Ethernet wired NICs.
- **USB.** These NICs provide network connections through a device plugged into the USB port. USB NICs are ideal for portable computers (e.g., laptops) and for devices that don't already have built-in NICs.

Key Components of an NIC

Controller: This is the brains of the NIC. It's a processor that handles the complex processes of data formatting, buffering, and communication with the computer's operating system.

Transceiver: This component is responsible for converting the digital data from the computer into signals suitable for the network medium (e.g., electrical signals for an Ethernet cable or radio waves for Wi-Fi). It also does the reverse, converting incoming signals back into digital data.

Memory (RAM/Buffer): NICs contain a small amount of memory to temporarily store incoming and outgoing data. This buffer helps manage the flow of data and compensates for speed differences between the network and the computer.

Bus Interface: The NIC uses this interface (like PCI or PCIe slots) to connect to the computer's motherboard, allowing it to communicate with the CPU and system memory.

MAC Address: This is a globally unique hardware address that's permanently embedded into the NIC's firmware by the manufacturer.

- A **diskless device** is a computer or workstation that operates without a local hard drive or solid-state drive (SSD). Instead, it loads its operating system and all necessary software from a central server over a network.
- A **Boot ROM** (Read-Only Memory) is a small, specialized chip or part of a chip on a Network Interface Card (NIC) that contains the initial instructions for a diskless device. This memory holds the code necessary to connect to a network and begin the process of loading the operating system from a server.
- **LED indicator.** Most NICs have an LED indicator integrated into the connector to notify the user when the network connects and data transmission occurs. Simply put, the indicator shows the NIC's working status (connected or not connected to the network).
- **NIC interface port.** This provides the physical connection point between the NIC and the network to exchange data signals, often using an Ethernet cable or transceiver.
- **Router.** Sometimes a router is needed to enable communication between a computer and other devices. In this case, the NIC connects to the router, which is connected to the internet.
- **Bracket for expansion slots.** NICs that need to be placed into a device's expansion slot require a profile bracket. Brackets are available in both compact and standard sizes.

Key NIC features

These features can differentiate one NIC from another:

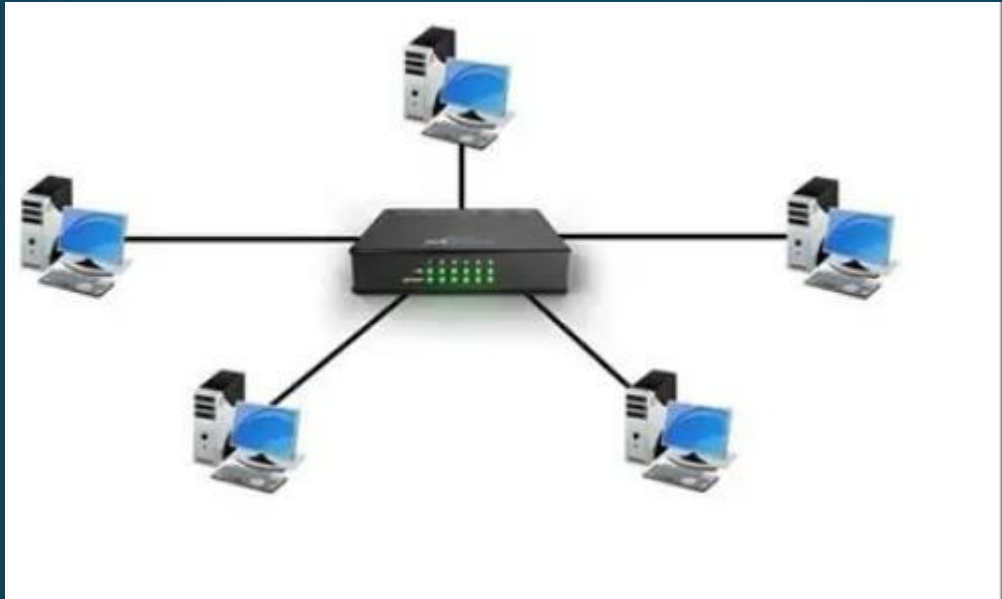
- **Speed.** The NIC speed refers to how fast data can travel over a network. All NICs have a speed rating -- listed in megabits per second (Mbps) or gigabits per second (Gbps) -- that determines the card's performance in a network. If the network's bandwidth is lower than the NIC or multiple computers connect with the same controller, the labeled speed decreases. Ethernet NICs commonly come in 10 Mbps, 100 Mbps, 1 Gbps, and 10 Gbps varieties. Higher-speed NICs (40 Gbps and above) are also available for high-end data transfer and high-performance computing tasks.

- **MAC address.** A unique, unchangeable MAC address, also known as a physical network address, is assigned to each NIC. MAC addresses deliver Ethernet packets to the intended endpoint.
- **Interrupt support.** Some NICs can trigger interrupts to notify the device CPU that an event has occurred (e.g., the device is receiving data). The interrupt is the CPU's signal to stop its current work and process the data. Modern NICs also use interrupt moderation, in which they wait for more packets to arrive before generating an interrupt, thus reducing the number of times the CPU gets interrupted.
- **Direct memory access (DMA).** Some NICs use DMA technology to access memory locations directly without involving the CPU. This allows the CPU to offload some of its data transfer tasks, enhancing system performance, and thus speeding up data movement and reducing latency. These NICs are often known as smartNICs.
- **Security features.** Modern NICs include numerous security features such as built-in firewalls, encryption capabilities and intrusion detection to protect data transmissions. Some NICs also use authentication protocols such as Password Authentication Protocol to authenticate users and prevent unauthorized or malicious users from accessing the network.

NETWORK DEVICES:

Network devices are physical devices that allow hardware on a computer network to communicate and interact with each other. Network devices like hubs, repeaters, bridges, switches, routers, gateways, and brouter help manage and direct data flow in a network. They ensure efficient communication between connected devices by controlling data transfer, boosting signals, and linking different networks.

Hub

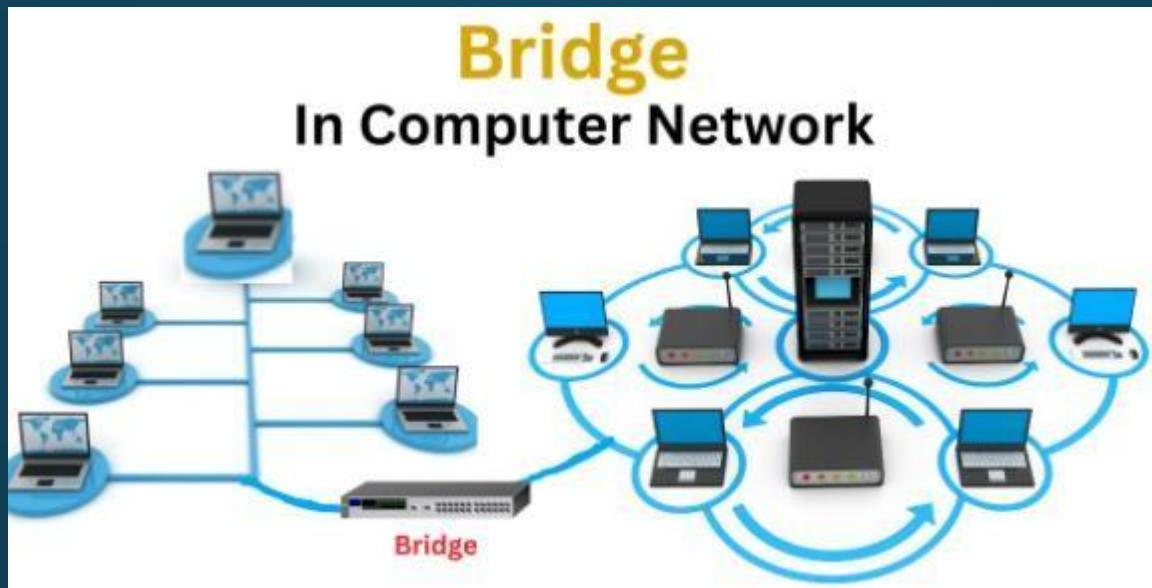


A hub is a multiport repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. In other words, the collision domain of all hosts connected through Hub remains one. Also, they do not have the intelligence to find out the best path for data packets which leads to inefficiencies and wastage.

Types of Hub

- **Active Hub:** These are the hubs that have their power supply and can clean, boost, and relay the signal along with the network. It serves both as a repeater as well as a wiring center. These are used to extend the maximum distance between nodes.
- **Passive Hub:** These are the hubs that collect wiring from nodes and power supply from the active hub. These hubs relay signals onto the network without cleaning and boosting them and can't be used to extend the distance between nodes.
- **Intelligent Hub:** It works like an active hub and includes remote management capabilities. They also provide flexible data rates to network devices. It also enables an administrator to monitor the traffic passing through the hub and to configure each port in the hub.

Bridge



A bridge operates at the data link layer. A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of the source and destination. It is also used for interconnecting two LANs working on the same protocol. It typically connects multiple network segments and each port is connected to different segment. A bridge is not strictly limited to two ports, it can have multiple ports to connect and manage multiple network segments. Modern multi-port bridges are often called Layer 2 switches because they perform similar functions.

Types of Bridges

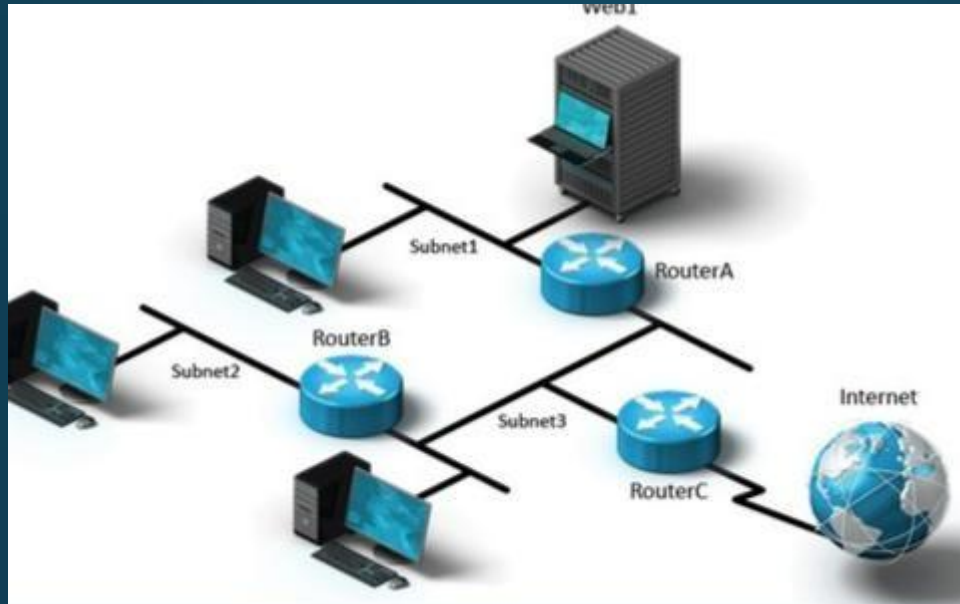
- **Transparent Bridges:** These are the bridge in which the stations are completely unaware of the bridge's existence i.e. whether or not a bridge is added or deleted from the network, reconfiguration of the stations is unnecessary. These bridges make use of two processes i.e. bridge forwarding and bridge learning.
- **Source Routing Bridges:** In these bridges, routing operations is performed by the source station and the frame specifies which route to follow. The host can discover the frame by sending a special frame called the discovery frame, which spreads through the entire network using all possible paths to the destination.

Switch

A switch is a multiport bridge with a buffer designed that can boost its efficiency(a large number of ports imply less traffic) and performance. A switch is a data link layer device. The switch can perform error checking before forwarding data, which makes it very efficient as it does not

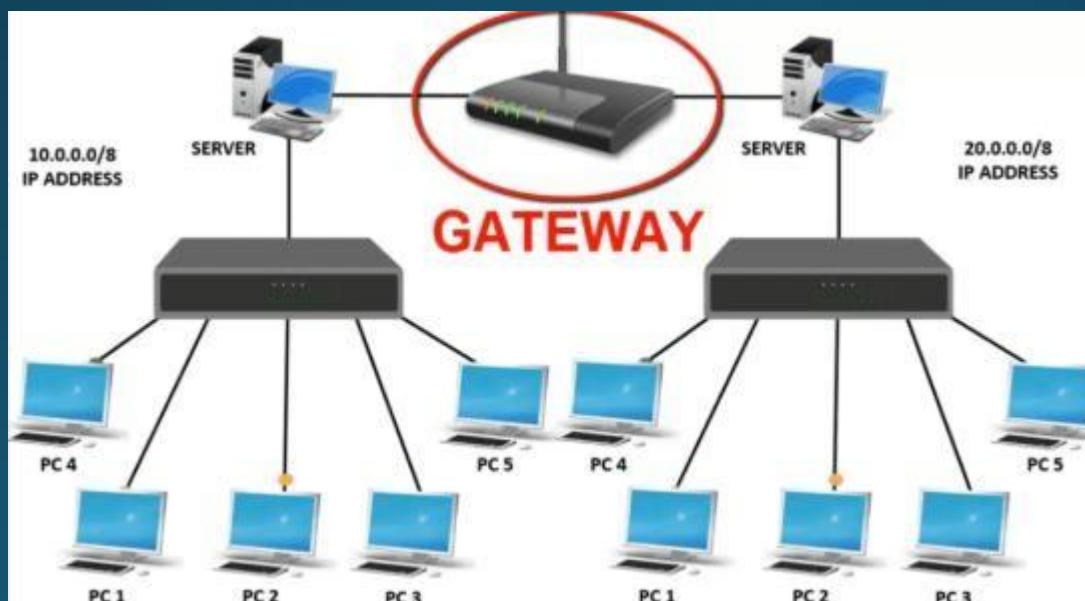
forward packets that have errors and forward good packets selectively to the correct port only. In other words, the switch divides the collision domain of hosts, but the broadcast domain remains the same.

Router



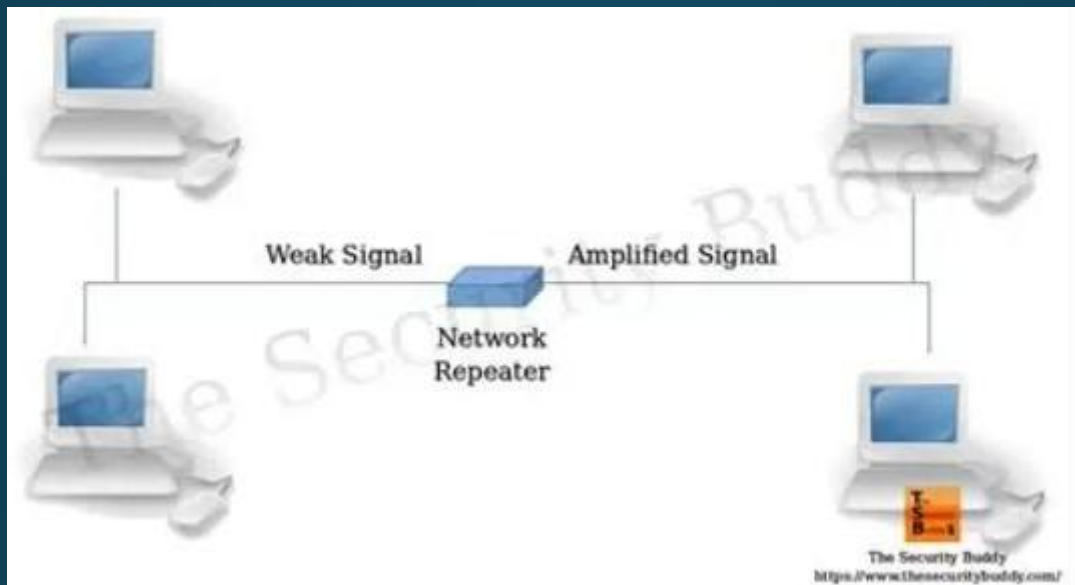
A router is a device like a switch that routes data packets based on their IP addresses. The router is mainly a Network Layer device. Routers normally connect LANs and WANs and have a dynamically updating routing table based on which they make decisions on routing the data packets. The router divides the broadcast domains of hosts connected through it.

Gateway



A gateway, as the name suggests, is a passage to connect two networks that may work upon different networking models. They work as messenger agents that take data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switches or routers.

Repeater



A repeater operates at the physical layer. Its main function is to amplify (i.e., regenerate) the signal over the same network before the signal becomes too weak or corrupted to extend the length to which the signal can be transmitted over the same network. When the signal becomes weak, they copy it bit by bit and regenerate it at its star topology connectors connecting following the original strength. It is a 2-port device.

A router and a gateway both move data between networks, but they operate on different principles and handle network protocols in fundamentally different ways. The key distinction is that a router works with a single type of network protocol, while a gateway is designed to translate between different protocols.

Router Protocol Usage A router's main job is to route traffic between networks that use the same protocol, most commonly TCP/IP. It makes decisions about the best path for data packets using routing protocols. These protocols are a set of rules routers follow to communicate with each other and share information about the network's topology.

Examples of Routing Protocols:

OSPF (Open Shortest Path First): A widely used protocol in large corporate networks. It allows routers to share information about the entire network and calculate the fastest route based on factors like bandwidth and network cost.

BGP (Border Gateway Protocol): The primary protocol that runs the internet. It is used by Internet Service Providers (ISPs) to exchange routing information between large, autonomous networks. BGP focuses on policy-based routing to ensure data traffic follows specific rules and paths.

RIP (Routing Information Protocol): An older, simpler protocol for small networks. It uses a "hop count" (the number of routers a packet must pass through) to find the shortest path.

Gateway Protocol Usage A gateway is a more complex device that acts as a translator between two networks that use different, incompatible protocols. Its function is to convert the data from one protocol into a format that the other network can understand, essentially bridging the communication gap.

Examples of Protocol Translation by Gateways:

Email Gateway: This type of gateway can translate emails from one protocol to another, such as converting a message from SMTP (used for sending email) to POP3 (used for retrieving it). This allows different email clients and servers to communicate seamlessly.

VoIP Gateway: These gateways are used in telecommunications to convert an analog telephone signal (from a traditional landline) into digital data packets that can be sent over a VoIP (Voice over IP) network. This allows traditional phones to connect to the internet.

IoT Gateway: These devices translate data from low-power, short-range protocols used by IoT sensors (e.g., MQTT or Zigbee) into standard internet protocols like TCP/IP, allowing the sensor data to be sent to the cloud for processing.

While a router ensures a packet gets from point A to point B on a network, a gateway's role is more fundamental: it ensures the two networks can even "talk" to each other in the first place by converting their communication languages. Your home router often contains a gateway to connect your local TCP/IP network to your ISP's network, which, from your perspective, is a completely different domain.

A bridge is a network device that connects and filters traffic between two or more network segments, effectively creating a single, larger network. It operates at the

Data Link Layer (Layer 2) of the OSI model. A bridge uses the MAC addresses of devices to intelligently forward data frames to the correct segment, helping to reduce network traffic and collisions.

---In simple terms, think of it this way:

A bridge is like an intersection with a traffic cop that directs cars only to the correct lanes on the same street. It ensures traffic stays organized and doesn't collide, but it doesn't send cars to a different city.

A router is like a highway interchange that directs traffic between different cities. It only works with cars (packets) that follow the same rules (protocols), like IP.

* A gateway is like a border checkpoint between two countries. It inspects everything, and if needed, translates the language and converts the documents (protocols) so that people (data) from one country can enter and be understood in the other.

Bridge vs. Router vs. Gateway

Feature	Bridge	Router	Gateway
OSI Layer	Data Link Layer (Layer 2)	Network Layer (Layer 3)	All layers, primarily layers 4–7
Primary Function	Connects and filters traffic within a single network .	Routes data packets between different networks that use the same protocol (e.g., TCP/IP).	Connects dissimilar networks and translates protocols between them.
Addressing	Uses MAC addresses to forward data frames.	Uses IP addresses to route data packets.	Uses both MAC and IP addresses and performs protocol translation.
Data Unit	Forwards frames .	Forwards packets .	Translates entire messages or data streams.
Examples	Connecting two different segments of a LAN to reduce collision domains.	A home Wi-Fi router connecting your home network to the internet.	An email server gateway converting an email from one protocol (e.g., SMTP) to another (e.g., POP3).

Feature	Repeater	Hub	Bridge	Switch	Router	Gateway
OSI Layer	1 (Physical)	1 (Physical)	2 (Data Link)	2 (Data Link)	3 (Network)	All layers
Feature	Repeater	Hub	Bridge	Switch	Router	Gateway
Main Function	Regenerates a weak signal to extend a network's range.	Broadcasts incoming data to all connected devices.	Connects two LAN segments and filters traffic.	Connects multiple devices and forwards data to the specific destination.	Connects different networks and routes data between them.	Connects networks with different protocols; acts as a protocol converter.
Data Unit	Bits	Bits	Frames	Frames	Packets	Varies (depends on the protocols being converted)
Traffic Direction	Bi-directional (replicates signal)	Bi-directional (broadcasts to all ports)	Filters and forwards based on MAC	Forwards intelligently to the specific destination	Routes intelligently based on IP addresses	Translates and routes between different

Featur e	Repeat er	Hub	Bridge	Switch	Route r	Gatew ay
			addre sses	ation port		protoc ols
Collisi on Domai n	Extend s a single collisio n domain .	A single, large collisio n domai n for all ports.	Divide s a collisi on domai n into smalle r segm ents.	Each port is its own collisio n domai n.	Each conne ction is its own broad cast domai n.	Varies
Addre ssing Metho d	None	None	MAC Addre ss	MAC Addre ss	IP Addre ss	Varies
Intellig ence	No intellig ence	No intellig ence	Some intellig ence (build s a MAC table)	Very intellig ent (builds a dynam ic MAC table)	Very intellig ent (build s a dyna mic routin g table)	Very intellig ent (proto col conver sion)